Decodability is not localization: auditing a hidden-state error probe

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Abstract

A linear probe that decodes "this reasoning has gone wrong" from hidden states is often read as a discovery about the model. We test how much of that reading survives contact with controls. In Qwen2.5-Math-1.5B-Instruct reading 3,360 ProcessBench traces, a probe on the selected layer predicts the persistent *invalid-so-far* target with held-out AUROC 0.866 $[0.849, 0.884]$, and adding the hidden representation to an equally tuned text-and-metadata model raises AUROC by 0.0587 $[0.0414, 0.0745]$. Three stronger claims fail. The score ranks boundaries within an erroneous trace almost perfectly (mean within-trace AUROC 0.968) but its threshold names the first wrong step in 28.9% of traces, fires 0.59 steps late on average, and fires on 38.8% of fully correct traces. Ranking transfers across the four ProcessBench sources (off-diagonal AUROC ≥ 0.739) while the thresholded localization metric does not (0.055 to 0.371). The causal test never runs: the model's own error verdict is below chance on the same boundaries (AUROC 0.342), so the steering results are uninterpretable rather than null. None of the three failures is visible in the pooled AUROC that a probe result usually reports, and we name the check that catches each one.

1 Introduction

Interpretability is increasingly used as an instrument of discovery. A probe decodes some property from hidden states, and the reported finding is a fact about the model: it represents this, it tracks that. "The model represents whether its own reasoning has become invalid" is a claim of that shape, and it is an appealing one. It promises a cheap step-level verifier and a story about self-monitoring.

A probe licenses less. It shows that a label is linearly decodable from a representation, under a chosen target, split, layer, and threshold. Four things do not follow, and each is a separate empirical question:

1. **Incremental information.** The label may be equally decodable from the visible text and from trace metadata, in which case the hidden state adds nothing.
2. **Localization.** A score that separates pre-error from post-error boundaries need not cross its threshold *at* the error. With a persistent label, every boundary after the error is positive, so accumulated evidence and position are enough to score well.
3. **Portability of the decision rule.** Ranking and thresholded detection can transfer at different rates across data sources.
4. **Causal use.** Decodable is not used (Hewitt and Liang, 2019; Belinkov, 2022; Elazar et al., 2021). Testing use requires a behavioral readout that works before it is perturbed.

We ran all four checks on one system: Qwen2.5-Math-1.5B-Instruct (Yang et al., 2024) reading 3,360 human-annotated mathematical reasoning traces from ProcessBench (Zheng et al., 2025). Check 1 passes. Checks 2 and 3 fail in specific ways. Check 4 cannot be run at all, because the

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model's own verdict about its reasoning is worse than chance on the boundaries we would have intervened on, and we report that as a failed precondition rather than as a null effect.

The point of the exercise is not the model. It is that all four failures are compatible with the headline number. Pooled AUROC of 0.866 is reported the same way whether or not a text baseline reaches 0.810, whether the threshold fires early or late, and whether the model's behavior responds to the direction. An interpretability result read as a discovery needs the checks that separate those cases. Here they all run on cached activations and stored predictions, on CPU, and they change what the study is allowed to say.

2 Related work

Process supervision scores intermediate reasoning steps rather than final answers (Cobbe et al., 2021; Lightman et al., 2023; Li et al., 2023). ProcessBench annotates the earliest erroneous step in traces drawn from GSM8K, MATH, OlympiadBench, and Omni-MATH (Zheng et al., 2025; Hendrycks et al., 2021; He et al., 2024; Gao et al., 2024).

Several recent studies score reasoning from internal states. Sun et al. (2025) probe arithmetic errors and re-prompt selectively on the result. ReProbe uses hidden-state step scoring for test-time scaling and STEP prunes traces with a hidden-state scorer (Ni et al., 2026; Liang et al., 2026). Alvarez and Baheri (2026) study first-error detection with hidden-state transport geometry across models and datasets. Yuan et al. (2026) separate diagnostic error awareness from causal use across several models. Our scope is narrower than any of these and our target is different: we hold the model and benchmark fixed and vary the evidential standard, so that the gap between a decoding result and a localization or causal result is measured on one set of predictions rather than argued.

3 Setup

Data and split. Each ProcessBench trace has a problem, a generator identity, a source, written steps, final-answer correctness, and a human first-error index e_i, with $e_i = -1$ for a fully correct trace. Traces sharing a normalized problem statement go to the same partition; seed 42 gives 60/20/20 train, validation, and test partitions. Of 3,400 attempted traces we retain 3,360 and 24,909 step boundaries; the 40 exclusions exceed the 2,048-token context limit. The test partition holds 669 traces (432 erroneous, 237 correct) and 4,985 boundaries. The model reads each fixed trace in one causal forward pass and does not generate it. Appendix A gives the full accounting.

Target and probe. At boundary k of trace i the target is $y_{ik} = \mathbf{1}[e_i \geq 0 \ \wedge \ k \geq e_i]$: the prefix is invalid so far. It marks the first invalid step and every later boundary, and it does not say whether the current step is locally wrong. Extraction records the 1,536-dimensional residual-stream state at each step boundary at 29 hidden-state indices. At each index we fit a standardized L2 logistic probe with class balancing over $C \in \{0.01, 0.1, 1, 10\}$. Validation AUROC selects index 23 and $C = 0.01$; validation Process F1 selects threshold 0.645 on a grid from 0.05 to 0.95. The probe is refit on train plus validation and evaluated once on test. The predicted first error is the first threshold crossing, or -1 if the score never crosses. Intervals are whole-trace bootstraps (1,000 draws for the probe and the nested comparison, 2,000 for the control audit).

Controls. Every nuisance model gets the probe's selection budget: the same C grid, a validation-selected threshold, a refit on train plus validation, and inverse boundary-count weights so each trace contributes equal training loss. Prefix TF-IDF sees the problem and reasoning text through the current boundary. Structural metadata sees source, generator, absolute and fractional position, trace length, and token count. A post-hoc semantic control encodes the same visible prefix with mean-pooled `all-MiniLM-L6-v2` embeddings under the same fitting procedure; its saved record does not pin an encoder revision or keep paired bootstrap draws, so we report point estimates. Final-answer correctness is a benchmark annotation that no online detector would have, so models using it are labeled diagnostic and never treated as baselines.

Post-hoc analyses. The temporal, boundary-location, transition, and semantic-baseline analyses were designed after the primary test result was known. We mark them post-hoc throughout. Their intervals describe resampling uncertainty inside the completed pipeline, not sensitivity to a new split or selection procedure.

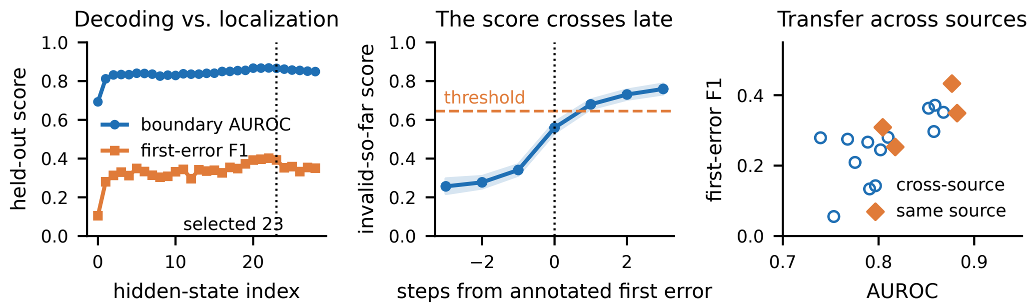

Figure 1: Left: held-out boundary AUROC and thresholded first-error F1 at each hidden-state index. The gap is present at every index, including the validation-selected one. Center: mean probe score aligned to the annotated first error, with a 95% whole-trace bootstrap band. At the onset the mean score is 0.559, below the operating threshold of 0.645; it crosses one step later. Right: the 16 source-pair evaluations. AUROC stays above 0.739 off the diagonal while first-error F1 falls as low as 0.055.

Table 1: Held-out comparison on 669 test traces. Every control receives the probe's selection budget and trace-equal training weights. Rows marked † use final-answer correctness, a benchmark annotation unavailable to an online detector; they are diagnostic, not baselines.

Predictor	AUROC	Proc. F1	Error exact	Corr. rej.
Hidden state (selected layer)	.866	.393	.289	.612
Prefix TF-IDF	.751	.274	.192	.477
Prefix MiniLM embeddings (post-hoc)	.753	.268	–	–
Structural metadata	.776	.168	.148	.194
Text + metadata	.810	.210	.169	.278
Metadata + final answer†	.854	.262	.164	.646
Text + metadata + final answer†	.874	.294	.176	.895

In the separately fit nested comparison (Appendix C), adding the selected-layer hidden representation raises AUROC by .0587 [.0414, .0745] over text plus metadata and by .0348 [.0242, .0451] over text, metadata, and final answer.

4 Results

4.1 Check 1: hidden states add information the visible text does not

The selected probe reaches test AUROC 0.866 $[0.849, 0.884]$, average precision 0.843 $[0.813, 0.870]$, and step F1 0.743 $[0.716, 0.769]$. The equally tuned controls are not far behind: 0.751 for prefix TF-IDF, 0.753 for the MiniLM prefix encoder, 0.776 for structural metadata, and 0.810 for text and metadata jointly (Table 1). A model given metadata plus final-answer correctness reaches 0.854, and one given text, metadata, and final-answer correctness reaches 0.874, above the probe. That field is an annotation, not a signal, but it shows how much of this target a benchmark's own bookkeeping predicts.

The nested comparison is the direct test. Let N be prefix text plus metadata and $N + H$ the same features plus the full selected-layer hidden representation. Adding H improves AUROC by 0.0587 $[0.0414, 0.0745]$, log loss by -0.0798 $[-0.1038, -0.0548]$, and Process F1 by $+0.1866$ $[0.1397, 0.2340]$. Adding H on top of the diagnostic feature set that already contains final-answer correctness still improves AUROC by 0.0348 $[0.0242, 0.0451]$. Both intervals exclude zero on this split. Check 1 passes, with the size of the gain worth noting: a tuned bag of words and six pieces of trace metadata recover most of what the probe recovers.

4.2 Check 2: the same score does not localize the first error

Within an erroneous trace, the probe orders boundaries almost perfectly: mean within-trace AUROC is 0.968 $[0.958, 0.976]$ over the 380 test traces whose first error is not at step 0. Any score that rises with step index would rank well on that measure, so it is not evidence about the representation. It is the ceiling that the decision rule built from the same scores fails to reach. That rule names the exact first-error step in 28.9% of erroneous traces, lands within one step in 59.3%, and within two in 72.7%. It crosses at all on 88.7% of erroneous traces and also crosses on 38.8% of fully correct ones. Among detected traces the mean signed error is $+0.59$ steps $[+0.36, +0.80]$: the crossing comes after the annotation, not before it.

Figure 1 (center) shows why. The mean score rises from 0.341 one step before the annotated error to 0.559 at the onset, 0.680 one step later, and 0.731 two steps later. At the onset the average trace is still below the operating threshold. The signal is a ramp, not an event, and a fixed threshold on a ramp fires late.

The alignment to the annotation is nonetheless real. Circularly shifting each frozen score trajectory 5,000 times, which preserves its values and destroys its alignment, gives a null exact rate of 0.164 $[0.137, 0.192]$ against the observed 0.289 ($p = 0.0002$), and a null Process F1 of 0.259 against the observed 0.393 ($p = 0.0002$). Comparing the score jump at a noninitial first error with the jump at a transition in a correct trace matched on source, generator, relative position, trace length, and token count gives 0.257 against 0.113, a paired difference of 0.144 $[0.096, 0.193]$. Correct traces drift upward too, which is the same ramp seen without an error to explain it.

Two robustness checks are flat. Choosing thresholds per length bucket moves Process F1 from 0.393 to 0.377 and correct-trace rejection from 0.612 to 0.540, leaving exact localization unchanged. Reading states at the last natural token of each step rather than at the artificial step marker changes AUROC by $+0.002$ $[-0.007, +0.011]$ and Process F1 by $+0.026$ $[-0.020, +0.075]$, so the result is not an artifact of the boundary token.

4.3 Check 3: ranking transfers, the operating point does not

Training a probe on one ProcessBench source and testing on another gives mean off-diagonal AUROC 0.805 against mean diagonal 0.845, with every off-diagonal value above 0.739. The thresholded metric moves much more: mean off-diagonal Process F1 is 0.261 against 0.336 on the diagonal, with off-diagonal values from 0.055 to 0.371 (Figure 1, right). The worst cell, GSM8K to OlympiadBench, keeps AUROC 0.753 while Process F1 falls to 0.055: the threshold fitted on short grade-school traces fires on 97% of correct OlympiadBench traces. These are four sources inside one benchmark construction, not four datasets, and they say nothing about another model family or size.

4.4 Check 4: the causal test does not run

To ask whether the model uses the decoded direction we need a behavioral readout that distinguishes valid from invalid prefixes before any intervention. We asked the model whether its reasoning contains an error and scored the difference between the conditional probabilities of "Yes" and "No". On 256 balanced boundaries this readout has AUROC 0.342 and specificity 0.203: worse than chance at the task the intervention would have measured. An earlier single-token CORRECT versus INCORRECT readout was worse, at AUROC 0.283 and specificity zero.

The pipeline gates the causal analysis on that readout clearing AUROC 0.5 with nonzero specificity, so the intervention is reported as an assay diagnostic. We moved the boundary state along the probe direction at $\alpha \in \{-4, -2, -1, 0, 1, 2, 4\}$ projection standard deviations and also along 20 random orthogonal directions at the extreme doses; every learned-direction interval includes zero, with a smallest detectable effect of 0.0029 at 80% power. None of that is evidence about the model. A flat dose-response curve on a broken instrument is uninformative, and reporting it as a null result would be the mistake the gate exists to prevent.

An exploratory positive. A probe on the onset difference $h_{i,e_i} - h_{i,e_i-1}$ separates error onsets from matched correct transitions with AUROC 0.769 $[0.713, 0.820]$ on 380 test pairs, above position (0.631), destination-step TF-IDF (0.671), and shuffled labels (0.585). The matching is imperfect: one placebo trace supports up to 46 pairs, and standardized differences are 0.146 for trace length and 0.285 for token count. The test split had been inspected before this analysis was conceived. We report it as exploratory and draw no mechanism claim from it.

5 What this changes about reading probe results

Four things generalize past this model, and each is cheap to check.

Report the decision rule, not the ranking. Within-trace AUROC of 0.968 and exact localization of 28.9% come from the same predictions. If the discovery claim is that a model knows *when* its reasoning went wrong, AUROC is the wrong summary, because it never touches the threshold that would answer the question.

Give the nuisance model the probe's tuning budget. Text plus metadata reaches 0.810 against the probe's 0.866. Measured against prefix TF-IDF alone the gap is 0.115, about twice the 0.0587 that survives the joint nuisance model, and measured against no control at all the reader sees 0.866 against chance.

Watch for benchmark bookkeeping in the feature set. Adding final-answer correctness, a field that sits in the ProcessBench metadata, lifts the nuisance model to AUROC 0.874, past the probe, while being unavailable to any detector running before the answer exists. Fields like this are easy to include by accident and hard to notice afterward.

Gate causal claims on assay validity. We could have run the steering experiment, found flat curves, and written that the direction is decoded but unused. The readout was below chance, so that sentence would have had no support. Checking the instrument before interpreting its output turned a publishable null into an honest gap.

6 Limitations

One model, one benchmark construction, one split seed. The nested gain in Table 1 is conditional on that partition; we have not repeated the grouped split, layer and C selection, and threshold selection across seeds, which is the first thing we would run next and needs no new extraction. The persistent target labels every boundary after the annotated error, so pooled metrics reward accumulated evidence. The MiniLM control is one unpaired post-hoc run with an unpinned encoder revision and does not exhaust visible-text representations. Calibration is not evaluated as a separate question; we report log loss and Brier score but no reliability analysis. The transition result is exploratory for the reasons given above, and the planned inverse-reuse and one-to-one matching refits have no completed artifact. Counterfactual activation patching produced no result: its corrections were never independently verified and it inherits the same failed behavioral gate.

Responsible use statement

The probe studied here is not a grader. At its selected threshold it flags 38.8% of fully correct solutions as containing an error and names the wrong step in most of the traces where an error exists, so using it for scoring, certification, or unattended feedback on student or user work would produce frequent and unevenly distributed false accusations; held-out correct-trace rejection already varies from 0.463 on OlympiadBench to 0.786 on GSM8K. The supported use is as a research diagnostic alongside human or formal verification. The same point applies past this probe: a score presented as evidence that a model "knows" it is wrong can be used to justify automation that the underlying evidence does not support, and the controls in Table 1 and the checks in Section 5 exist to make that gap visible. The data are model-generated mathematical solutions from a public benchmark and contain no personal information. The analysis package, configuration, frozen prediction artifacts, and figure code that reproduce every number here are included in the anonymized supplement; extraction needs one A100-class GPU and the remaining analyses run on CPU.

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A Data accounting

Retained boundary counts by source, in the order shown above, are 1,229 / 3,718 / 5,129 / 4,868 for train, 414 / 1,303 / 1,708 / 1,555 for validation, and 439 / 1,387 / 1,661 / 1,498 for test. The attempted total was 25,697 boundaries and the retained total is 24,909.

Table 2: Attempted and retained traces and boundaries by frozen partition.

Partition	Attempted	Retained	Excluded	Boundaries	Error traces	Correct traces
Train	2,041	2,014	27	14,944	1,311	703
Validation	681	677	4	4,980	444	233
Test	678	669	9	4,985	432	237
All	3,400	3,360	40	24,909	2,187	1,173

Table 3: Attempted/retained/excluded traces by source and frozen partition.

Partition	GSM8K	MATH	OlympiadBench	Omni-MATH
Train	236/236/0	597/592/5	602/594/8	606/592/14
Validation	80/80/0	198/198/0	204/202/2	199/197/2
Test	84/84/0	205/204/1	194/188/6	195/193/2

B Full metric table and subgroup audit

The selected probe's held-out metrics are AUROC 0.866 $[0.849, 0.884]$, average precision 0.843 $[0.813, 0.870]$, step F1 0.743 $[0.716, 0.769]$, error exactness 0.289, correct-trace rejection 0.612, Process F1 0.393, and complete-trace accuracy 0.404. Weighting traces equally at evaluation rather than by boundary count changes AUROC by $+0.0013$ $[-0.0069, 0.0103]$, average precision by -0.0052, and step F1 by -0.0087, all with intervals covering zero.

Table 4: Trace-level outcomes by ProcessBench source, with 95% whole-trace bootstrap intervals. These are descriptive subgroup results, not independent hypothesis tests.

Source	Complete accuracy	Exact error	Correct rejection
GSM8K	.512 [.405,.619]	.238 [.116,.375]	.786 [.658,.902]
MATH	.500 [.426,.569]	.347 [.264,.435]	.723 [.620,.818]
OlympiadBench	.298 [.234,.362]	.207 [.138,.282]	.463 [.340,.585]
Omni-MATH	.358 [.290,.425]	.324 [.253,.401]	.467 [.319,.614]

Complete-trace accuracy is 0.292 $[0.245, 0.342]$ among traces with an incorrect final answer and 0.506 $[0.451, 0.554]$ among traces with a correct one. Across generators contributing at least 20 held-out traces it ranges from 0.316 to 0.621.

C Nested comparison and intervention records

D Reproduction record

The package exposes one configuration file and one command group:

```
uv sync --extra dev
uv run math-error --config configs/project.yaml validate-config
uv run math-error --config configs/project.yaml run-all
uv run math-error --config configs/project.yaml analyze
uv run math-error --config configs/project.yaml fit-conditional
uv run math-error --config configs/project.yaml fit-contextual-baseline
uv run math-error --config configs/project.yaml fit-transition
uv run math-error --config configs/project.yaml transition-diagnostics
uv run math-error --config configs/project.yaml analyze-boundaries
uv run pytest && uv run ruff check src tests
```

The configuration records seed 42, model and dataset identities, the 2,048-token limit, selection grids, bootstrap counts, and artifact paths. Extraction ran on one NVIDIA A100 in bfloat16 at batch size 16 over single causal passes; the extraction record fixes the model identity, dtype, token limit, and dataset SHA-256. Probe fitting and every audit in this paper run on CPU, and the audit record stores Python, NumPy, pandas, SciPy, scikit-learn, matplotlib, and platform versions. Dataset and model use follows the licenses of the published sources.

Table 5: Nested feature-set comparison. N is prefix TF-IDF plus structural metadata; O adds final-answer correctness; H is the full selected-layer hidden representation. All conditions share the partitions, the C grid, trace-equal training weights, and the refit protocol.

Condition	AUROC	Log loss	Proc. F1	Δ AUROC vs. nuisance only
N	.811	.533	.211	
$N+H$	.869	.454	.397	+.0587 [.0414, .0745]
O	.874	.440	.284	
$O+H$	.909	.385	.444	+.0348 [.0242, .0451]
H alone	.868	.457	.397	

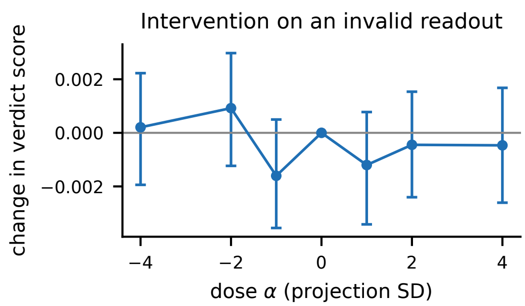

Figure 2: Paired change in the model's verdict score under the learned-direction intervention at each dose, with 95% intervals. Every interval covers zero. Because the unperturbed readout has AUROC 0.342, this figure records an assay diagnostic and carries no evidence about causal use.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Introduction name the single model and benchmark, distinguish the persistent prefix target from onset detection, and state the negative and post-hoc results.

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   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 6 discusses one-model and one-split scope, shortcut sensitivity, the post-hoc status of several analyses, the untested calibration question, and the invalid causal assay.

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